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CO_2/CH_4 Gas Separation by Polyoxometalate Ionic Liquid Confined in Carbon Nanotubes Using Molecular Dynamics Simulation

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Cite This: *Ind. Eng. Chem. Res.* 2023, 62, 14548−14556
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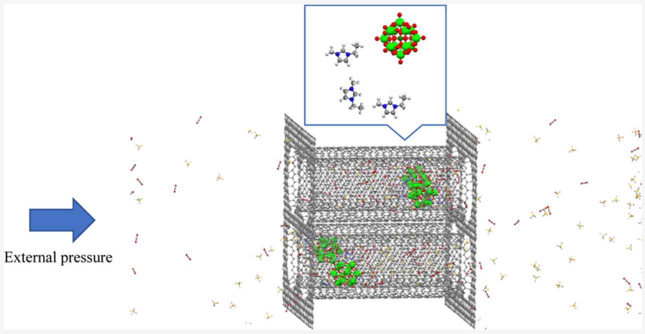
ABSTRACT: In this study, we have performed molecular dynamics (MD) simulations to investigate the separation of a CO_2/CH_4 gas mixture using 1-ethyl-3-methylimidazolium tungstophosphate ($[Emim]_3[PW_{12}O_{40}]$) or ($[Emim]_3[Keggin]$) ionic liquid confined into a carbon nanotube (CNT) at several external pressures at 300 K. We have also compared the results with the different ionic liquids (ILs), including imidazolium tetrafluoroborate ($[Emim][BF_4]$), imidazolium hexafluorophosphate ($[Emim][PF_6]$), and imidazolium tetrachloroaluminate ($[Emim][AlCl_4]$). In another model, the membrane was constructed by four similar CNTs (with the confined ILs), which are almost spaced equally from each other (4ILs−4CNTs). Our results indicated the importance of polyoxometalate IL−CNT nanocomposites in the gas separation process. CNTs with $[Emim]_3[PW_{12}O_{40}]$ IL exhibit the highest sorption of CO_2 owing to the high electric charge of this particular IL. It is also totally observed that $AlCl_4^-$ exhibits smaller sorption than the other ILs, especially for the 4ILs−4CNTs system for CO_2 sorption. The $[Emim]_3[PW_{12}O_{40}]$ IL in the 4ILs−4CNTs configuration exhibits more than 2 times rejection rate than the 4CNTs without IL. The diffusion coefficients of confined gas molecules in the CNT are also smallest for the [Keggin][Emim] IL, which are in agreement with the sorption and rejection results.



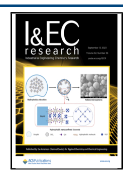
1. INTRODUCTION

Carbon dioxide (CO_2) and methane (CH_4) are both gases that can be found in various concentrations in the atmosphere and in industrial processes.[1,2] The separation of CO_2 and CH_4 is important because they have different uses and properties, and it is also important for reducing greenhouse gas emissions.[3,4] Separating CO_2 from other gases can be done through a variety of methods, including chemical sorption, membrane separation, and cryogenic distillation.[5−8] Chemical sorption is the most common method used to separate CO_2 from other gases, and it involves passing the gas mixture through a solution that absorbs the CO_2.[9−11]

Ionic liquids (ILs) are salts that are liquid at room temperature and have gained interest for their potential application in gas separation. They possess unique properties such as high thermal stability, low vapor pressure, and tunable selectivity, which make them attractive for gas separation processes.[7,12−14] The characteristics of ionic liquids are influenced by both their anion and cation. The ethylmethylimidazolium cation, also known as Emim, is a commonly used cation in ILs due to its superiority over other cations in several aspects such as thermal stability, good ion conductivity, commercial availability, and nontoxicity.[8,15] These advantages have led to widespread use of Emim cations in ILs for a range of applications, including energy storage, catalysis, and gas separation.[7,16] However, ILs are not commonly used alone for gas separation due to several limitations, including high cost, volatility, and lack of selectivity for specific gas mixtures. Additionally, ILs may not have sufficient mechanical stability for long-term use in gas separation systems.[17] To overcome these limitations, ionic liquids are often used in combination with other materials, such as polymers, zeolites, metal−organic frameworks (MOFs), and nanoporous carbons to enhance their performance in gas separation applications.[18,19] However, for CO_2 sorption, the sorbers must have strong stability at higher temperatures, strongly acidic conditions, and humidity. Unfortunately, most organic polymers lose their stability at high temperatures, and most MOFs are also unstable in aqueous solutions. Carbon nanotubes (CNTs) absorb CO_2 almost twice as much as activated carbons at 0−200 °C and showed higher selectivity for CO_2 in CO_2/CH_4 mixtures.[20−23]

The use of ionic liquid enclosed in CNTs has attracted much attention for gas separation due to several advantages, including

Received: June 8, 2023
Revised: August 18, 2023
Accepted: August 18, 2023
Published: September 5, 2023



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Ind. Eng. Chem. Res. 2023, 62, 14548−14556

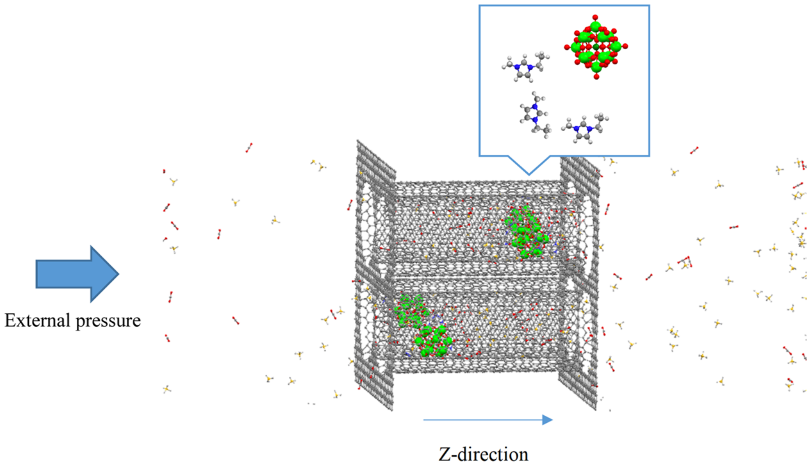

Figure 1. Snapshot of the 4ILs−4CNTs system (containing $[Emim]_3[PW_{12}O_{40}]$ IL) at P = 5 bar and T = 300 K. The C atoms are in gray (the C atoms of methane are in yellow), O atoms are in red, H atoms are in white, W atoms are in green, the P atom is in dark green, and N atoms are in blue.

high surface area, improved selectivity, thermal stability, high ionic conductivity, and scalability.[24,25]

Using molecular dynamics simulation, Tian et al. showed that CO_2, N_2, and CH_4 gases can penetrate through a nanopore in graphene without any selectivity. However, when a monolayer of ionic liquid [Emim][BF_4] is deposited on porous graphene, CO_2 has a much greater permeability than the other two gases. The anion dynamically adjusts the pore size by floating above the pore, and its affinity provides a complex for CO_2, while the larger cation (which cannot pass through the pore) holds the anion in place through electrostatic attraction.[26] Zeng et al. used molecular simulation to investigate the absorption and separation of CO_2/N_2 mixtures. They showed that the entry of ionic liquid into CNT significantly increases the absorption of CO_2 while being insensitive to the absorption of N_2. This is due to the strong interaction between the ionic liquid and CO_2 within the carbon nanotube.[27] Zeeshan et al. deposited 1-*N*-butyl-3-methylimidazolium hexafluorophosphate ([BMIM]-[PF_6]) on reduced graphene aerogels (rGAs). The results showed that there is interaction between the rGA levels and the ionic liquid, which significantly improves the CO_2 adsorption and increases the selectivity of CO_2/CH_4 adsorption.[28] Vicent-Luna et al. reported a molecular simulation study aimed to ascertain the effect exerted in gas adsorption when room-temperature ionic liquids (RTILs) are added into the pores of the Cu−BTC metal−organic framework (MOF). It is observed that the presence of RTILs in the MOF pores enhances CO_2 adsorption significantly at low pressures, whereas methane and nitrogen adsorption is unaffected.[19]

The polyoxometalates are nanoclusters of metal cations and oxoligands, which have significant chemical properties and have significant applications in materials science and catalysis.[29−31] The Keggin anion ($PW_{12}O_{40}$) is one of the most representative of the polyoxometalates which has focused several experiments and theoretical methods on its family.[32−34] In this study, we have performed molecular dynamics (MD) simulations to investigate the separation of the CO_2/CH_4 gas mixture using 1-ethyl-3-methylimidazolium tungstophosphate ($[Emim]_3[PW_{12}O_{40}]$) or ($[Emim]_3$[Keggin]) ionic liquid confined into CNTs at several external pressures at 300 K. We have also compared the results with the different ILs. To examine the results, we have calculated different properties, including the number of gases sorbed on the confined ILs, rejection rate, radial distribution function (RDF), and diffusion coefficient.

2. SIMULATION DETAILS

We conducted NVT-MD simulations on ILs, including 1-ethyl-3-methylimidazolium tungstophosphate ($[Emim]_3[PW_{12}O_{40}]$) or ($[Emim]_3$[Keggin]), imidazolium tetrafluoroborate ([Emim][BF_4]), imidazolium hexafluorophosphate ([Emim]-[PF_6]), and imidazolium tetrachloroaluminate ([Emim]-[$AlCl_4$]) confined within a CNT. The simulation was performed in a 170 × 60 × 60 $Å^3$ box designed using modified DLPOLY software[35] with the interior space enclosed by two graphene walls containing the membrane, which is an IL-filled zigzag (26,0) CNT of 50 Å length fixed in its position (Figure 1). Such hybrid structures of graphene pillared with CNTs that have been found to be theoretically stable and can be constructed experimentally seem truly promising materials serving as gas separation membranes.[36−38] They combine adsorptive and transport properties of both species, leading to a stable, chemically uniform 3-D network.[38] On the left side of this membrane, we have positioned a mixture of CO_2/CH_4 gas with 150 molecules of each type. It has been shown that the rigid Zhang force field[39] generally performed best among the different CO_2 force fields,[40] providing viscosities and self-diffusion coefficients. This model has been proven to provide the best overall representation of CO_2 in molecular simulation studies over a wide range of thermodynamic conditions.[40] The OPLS CH_4 model[41] was shown to give the best overall estimates of viscosities and self-diffusion coefficients with quite good accuracy. Therefore, we have used the Zhang and OPLS force fields for CO_2 and CH_4, respectively. The [Emim] cation was modeled using the OPLS-AA potential.[42] The Keggin anion was modeled as a rigid particle, and the all-atom force field parameters of the $PW_{12}O_{40}$ anion were taken from the work by López et al.,[43] which is based on the Amber force field and has been extensively applied for the polyoxometalate anion simulation.[44,45] The anions [PF_6], [BF_4], and [$AlCl_4$] were simulated as rigid molecules.[42] The use of rigid anions in OPLS-AA has been shown to provide an accurate representation of the ionic liquid's physical properties,[46] including its use as a reaction medium for a computed QM/MM Diels−Alder reaction study.[47] The OPLS-AA model is a well-known force field which has been used successfully to simulate different ILs in

https://doi.org/10.1021/acs.iecr.3c01924
Ind. Eng. Chem. Res. 2023, 62, 14548−14556

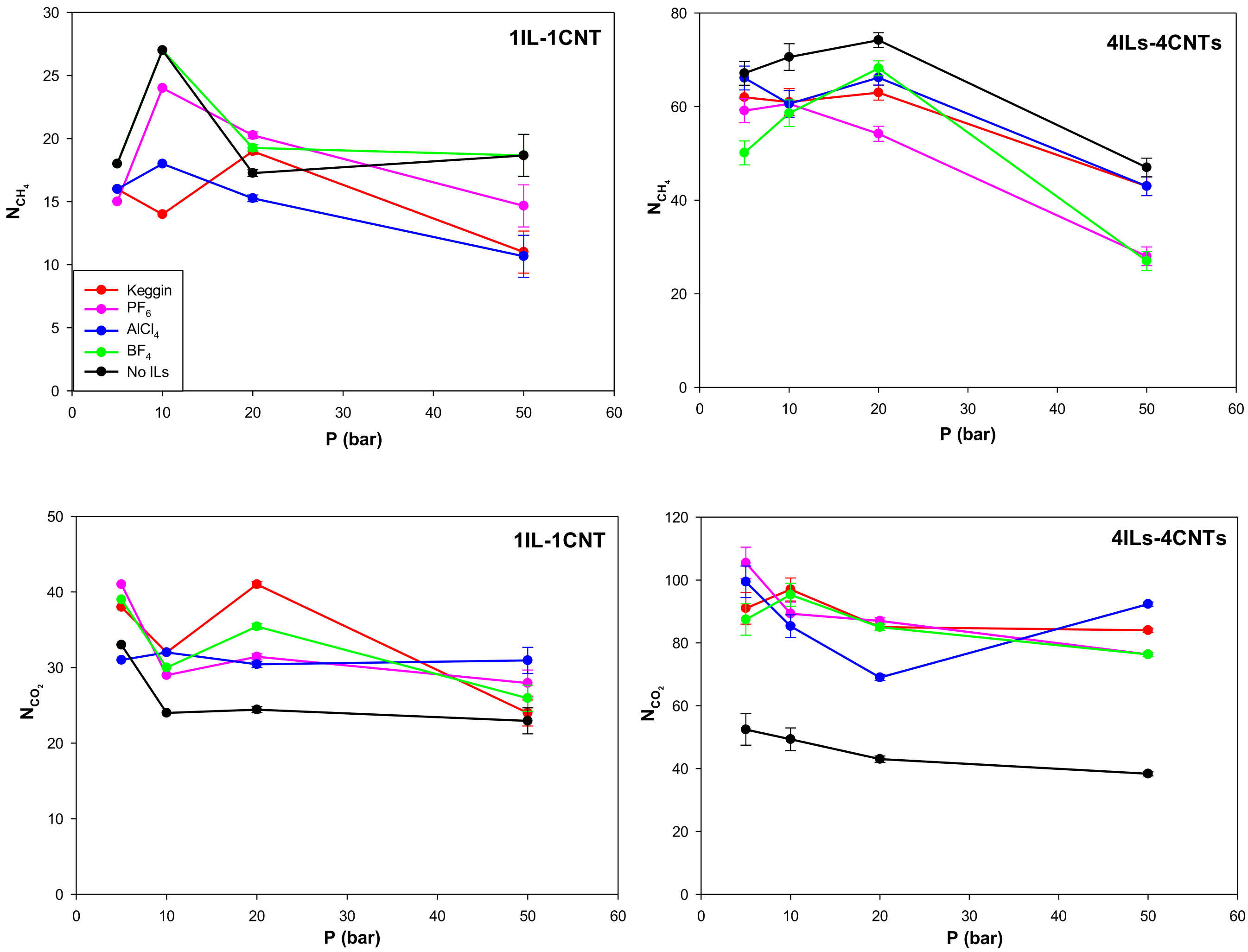

Figure 2. Number of sorbed CH_4 and CO_2 molecules in CNT−IL systems at 300 K at different pressures.

nanoscales in the recent years.[42,46,48−52] As a brief overview, the OPLS-AA model evaluates the total energy of the system as a sum of individual energies for the harmonic bond stretching and angle bending terms, a Fourier series for torsional energetics, and Coulomb and 12−6 Lennard-Jones (LJ) terms for the nonbonded interactions.[42] The LJ model was also used for other remaining interactions. The parameters of all interactions in this work are presented in Table S1.

The coulombic long-range interactions were calculated using the mesh Ewald method with a precision of 10^{-6}. All interatomic interactions between the atoms in the simulation box have been calculated within a cutoff distance of 15 Å. Periodic boundary conditions have been applied in all dimensions, but to prevent the mixing of the separated gases with the ones that have not passed through, a graphene wall has been placed on the left side. All graphene walls are kept in a fixed position. In the first stage, an IL-content CNT is placed between two graphene walls. In the next stage, the number of CNTs has increased to four. These four CNTs are equally spaced and placed next to each other. The gas mixture was initially allowed to equilibrate for 1 ns at 300 K, and then a part of the graphene wall located opposite the two ends of the CNTs was removed and the whole system was simulated for 5 ns with the external pressures of 5, 10, 20, 50, and 100 bar (in addition to the initial pressure of the gas in the left side). To apply an external constant pressure to the gas molecules (from left to right in the z direction), Zhu et al.'s technique[53,54] was used. In this way, a constant force is exerted on both the CO_2 and CH_4 molecules, according to the following relation:

$$P\,(\mathrm{Pa}) = \frac{nF\,(\mathrm{N})}{A\,(\mathrm{m}^2)} \tag{1}$$

where n is the number of gas molecules, F is the value of induced constant force, P is the applied pressure, and A is the area of the box. This technique has been used in many simulations of pressure-driven flow,[55,56] including our previous works.[57,58]

3. RESULTS AND DISCUSSION

We have presented the snapshot of the 4ILs−4CNTs system containing the $[Emim]_3[PW_{12}O_{40}]$ IL at P = 5 bar and T = 300 K in Figure 1.

The number of sorbed CO_2 and CH_4 molecules using both one IL in one CNT (1IL−1CNT) and four ILs in four carbon nanotubes (4ILs−4CNTs) is shown in Figure 2. In general, the sorption of CO_2 in the IL−CNT systems is higher compared to CH_4. This is due to the stronger interaction of CO_2 molecules with the CNT and IL. Similar results have been reported in the CO_2/CH_4 mixture using different carbon nanomaterials by Akbarzadeh and Abbaspour,[59] Palmer et al.,[60] and Skarmoutsos et al.[61] The number of CO_2 molecules sorbed in the low-

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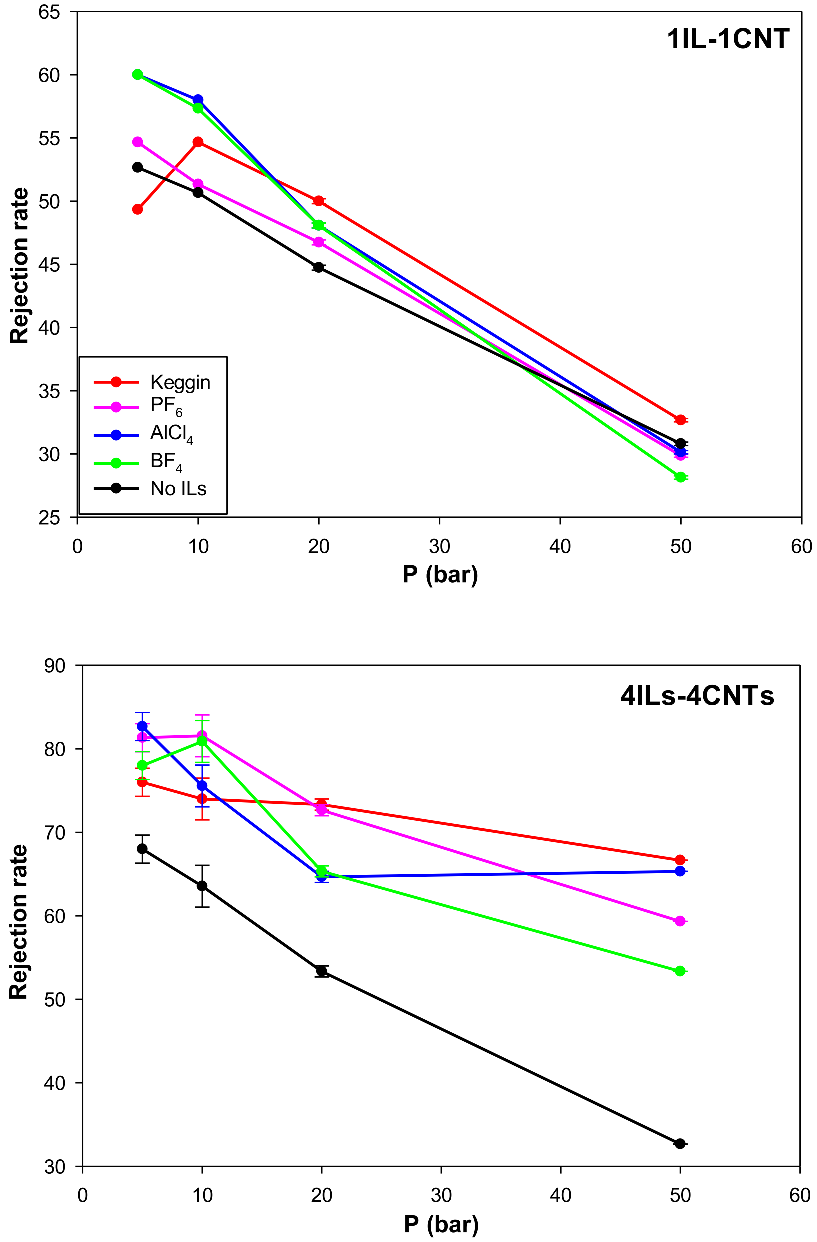

Figure 3. Rejection rate of CO_2 molecules in CNT−IL systems at 300 K at different pressures.

Table 1. Rejection Rate of CO_2 in the One CNT and Four CNTs Systems Including [Emim][Keggin] IL

	rejection rate of CO_2	
temperature (K)	1CNT	4CNTs
300	54.66	74.00
350	52.44	69.33

pressure region with different ILs and even without them is greater in four carbon nanotubes than in one carbon nanotube. This is because the sorption increases with an increase in the number of carbon nanotubes and ILs. It is also observed that the number of sorbed gases decreases by increasing the external pressure. This is due to the fact that the sorbate−sorbent interactions are very important at low pressures, and they are not significant at higher pressures. Similar results have also been achieved by Palmer et al.[60] for the adsorption of the CO_2/CH_4 mixture in a carbon nanospace using molecular simulations.

We can also observe a peak in the charts of CH_4 around 10 bar, which indicates a sudden increase in the number of sorbed CH_4 molecules. As CH_4 sorption begins to increase, the amount of CO_2 sorption decreases as a result of the sorption competition between the two species. The number of sorbed CH_4 molecules increases at lower pressures as a result of an increase in density inside the pore, which entropically favors CH_4 over CO_2 because of its spherical shape and efficient packing geometry.[59,60] As the pressure decreases, the external force leads to the decrease in the density inside the pore and results in the decrease of sorption of CH_4 over CO_2.

Carbon nanotubes with $[Emim]_3[PW_{12}O_{40}]$ IL exhibit the highest sorption of CO_2 (with the exception of highest pressure) owing to the high electric charge of this particular IL. Totally, the CNTs without IL exhibit higher sorption of CH_4 (especially in 4CNTs), which is due to the more accessible volume and gas density inside the empty CNT. Also, CO_2 with its smaller size has a higher passage rate compared to CH_4. For the same

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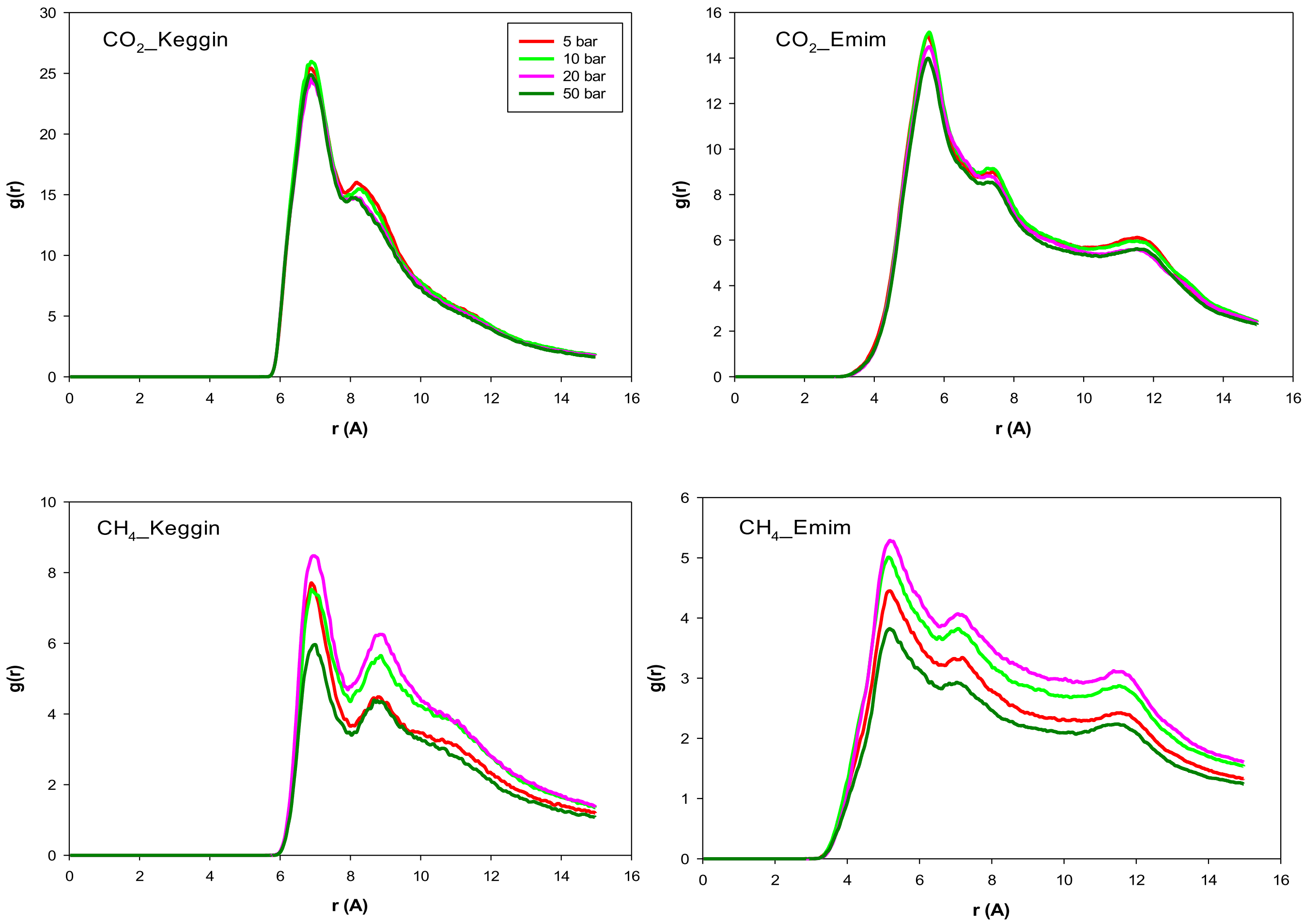

Figure 4. CO_2−Keggin, CO_2−Emim, CH_4−Keggin, and CH_4−Emim RDFs in 1CNT−1IL system at 300 K at different pressures.

reasons, the CNTs without IL exhibit smallest sorption of CO_2. We cannot observe an ordered trend for the other IL systems. However, it is also totally observed that the $AlCl_4$ anion exhibits smaller sorption than the other ILs, especially for the 4ILs−4CNTs system for CO_2 sorption. The charge of $AlCl_4$, PF_6, and BF_4 anions is the same. Therefore, this result can be due to the less accessible volume in the $AlCl_4$-containing CNT than that of PF_6 and BF_4 because the Al−Cl bond length is longer than the P−F and B−F bond lengths.

Given that CO_2 sorption is higher than CH_4 in the CNT−IL nanocomposites, the rejection of CO_2 is higher than that of CH_4. The rejection rate of CO_2 molecules is defined by the following equation:

$$R = \frac{N_F - N_P}{N_F} \times 100 \tag{2}$$

where R is the rejection rate of CO_2, N_F is the initial number of CO_2 molecules in the gas mixture on the left side (N_F = 150), and N_P is the number of permeated CO_2 molecules on the right side (Figure 1). We have presented the rejection rate of CO_2 molecules using different CNT−IL systems in Figure 3. As this figure shows, totally, the presence of the ILs in the 1CNT and 4CNTs configurations results in a greater CO_2 rejection rate, which is due to the IL−CO_2 interactions. With the exception of very low pressure (P = 5 bar), the rejection rate of $[Emim]_3[PW_{12}O_{40}]$ IL is higher than the other ILs. This result is in agreement with the gas sorption in Figure 2 and is due to the high electric charge of this particular IL. The nanocomposites show a higher rejection rate for CO_2 over CH_4 at low pressure ranges. As the pressure increases, the volume of gas entering the CNT increases, and with the saturation of the IL trapped inside the CNT, the rejection decreases. According to Figures 2 and 3, by increasing the number of CNTs, the gas sorption increases. Also, the rejection rate of CO_2 increases by increasing the CNTs. However, the increase in gas selectivity is not as much as the gas sorption by increasing the CNT. The $[Emim]_3[PW_{12}O_{40}]$ IL in the 4ILs−4CNTs configuration exhibits more than 2 times rejection rate than the 4CNTs without IL, which approves the importance of polyoxometalate IL−CNT nanocomposites in the gas separation process.

More gas separation investigations have been performed at 300 K.[59,62] Huang et al.[63] examined the effect of temperature on gas sorption and separation including CO_2 and CH_4 and found that the increase in temperature leads to the decrease in gas sorption and separation. For instance, we have also calculated the rejection rate of CO_2 gas in the one CNT and four CNTs systems including [Emim][Keggin] IL and confirmed this decrease (Table 1).

The decrease in the sorption and rejection rates is due to the weakening of the interactions by increasing the temperature.

To further investigate the structural properties of gas sorption in confined [Keggin][Emim] IL, which showed more gas sorption than the other ILs, the radial distribution functions (RDFs) of the center of mass pairs between CO_2 and CH_4 molecules and the Keggin anion and Emim cation in single CNT are presented in Figure 4. According to the work of Mendonça,[64]

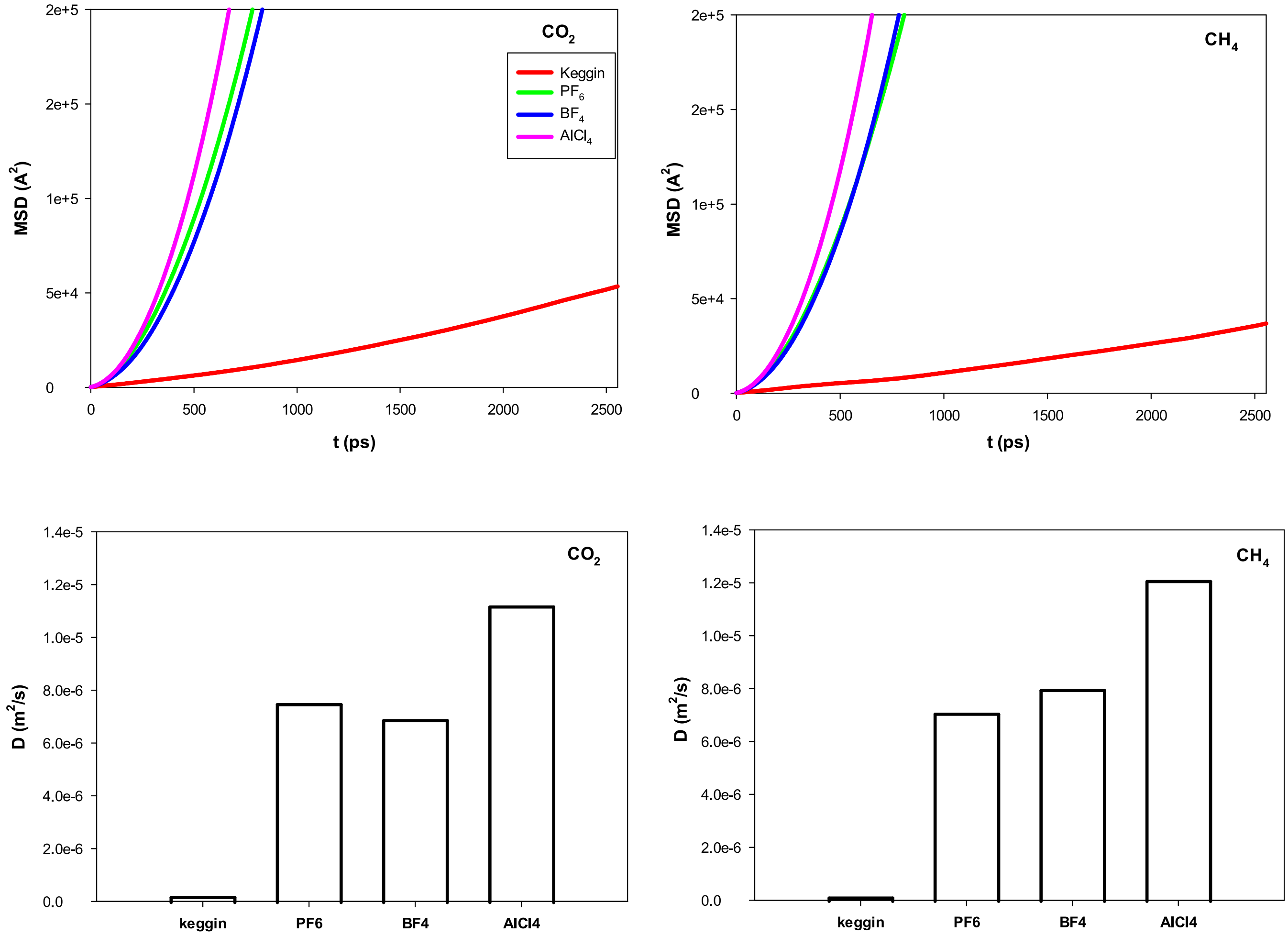

Figure 5. MSD and diffusion coefficients of confined gas molecules in the single CNT in different systems.

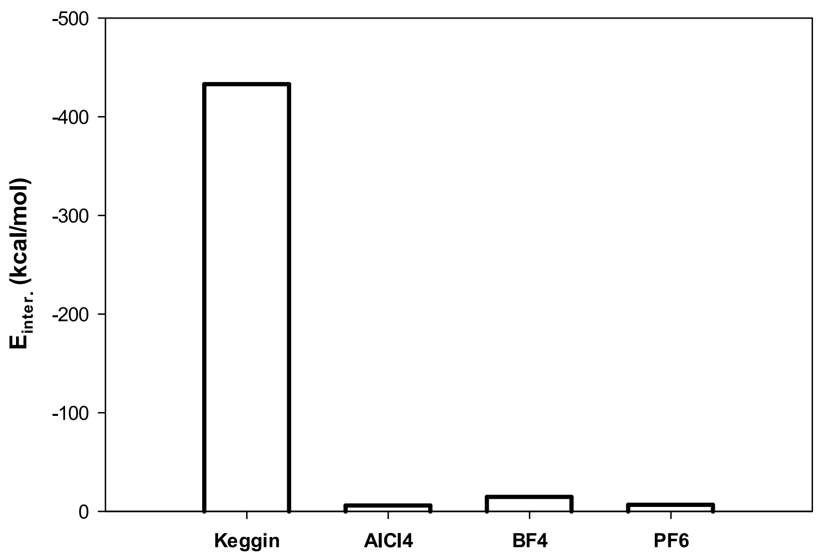

Figure 6. Interaction energy between the confined CO_2 and the different ILs.

the N atom has been selected as the representative of the center of mass of the Emim cation. The results clearly indicate that the first CO_2−Keggin and CO_2−Emim RDF peaks are much higher than those of CH_4, which approves the stronger CO_2−IL interactions than the CH_4−IL interactions. Due to the same reason, the CO_2−Keggin RDFs also appear at shorter distances than those of CH_4−Keggin. It is also shown that the first gas−anion RDF peak is higher than that of the gas−cation, which is due to the stronger gas−anion interactions owing to the greater electrostatic charges on the Keggin anion than the Emim cation. In agreement with the sorption results in Figure 2, it is shown

https://doi.org/10.1021/acs.iecr.3c01924
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that the RDF peaks decrease by increasing the external pressures.

To examine the dynamical properties of the confined gas molecules with different ILs in the nanosystems, the CNTs and their gases contained in different systems have been separated from the remaining systems and simulated for 4 ns in the NVT ensemble with an equal number of CO_2 and CH_4 gas molecules (the excess molecules were deleted). The diffusion coefficients of confined gas molecules in the single CNT have been computed from the slopes of mean square displacement (MSD) plots using the following equation[65] and are presented in Figure 5:

$$D = \lim_{t\to\infty} \frac{1}{6tN} \sum_{i=1,N} \langle |r_i(t) - r_i(0)|^2 \rangle \tag{3}$$

where N is the number of particles, and the positions of the particles at time $t = 0$ and t are shown by $r_i(0)$ and $r_i(t)$, respectively. According to Figure 5, the diffusion coefficients of confined gas molecules are smallest for the [Keggin][Emim] IL, which is in agreement with the sorption and rejection results in Figures 2 and 3 and is due to the greater gas−[Keggin][Emim] IL interactions. Also, the diffusion coefficients of confined CO_2 molecules are smaller than those of the CH_4 molecules, which is due to the stronger CO_2−IL interactions compared to CH_4−IL interactions. In agreement with the sorption results in Figure 2, $AlCl_4^-$ exhibits the highest diffusion coefficients (due to the smaller sorption on the confined IL in the CNT).

To approve the greater CO_2−IL interactions in the [Keggin][Emim] system confined in the CNT compared to the other confined systems, the interaction energy between the confined CO_2 and the different ILs have also been presented using the following relation:

$$E_{\text{inter.}} = E_{\text{CNT+CO}_2\text{+IL}} - (E_{\text{CNT+CO}_2} + E_{\text{CNT+IL}}) \tag{4}$$

where $E_{\text{inter.}}$, $E_{\text{CNT + CO}_2\text{ + IL}}$, $E_{\text{CNT + CO}_2}$, and $E_{\text{CNT + IL}}$ are the interaction, total (CNT + CO_2 + IL), confined CO_2, and confined IL molecule configurational energies, respectively. In these simulations, the CNT was fixed, and therefore its energy was neglected. As Figure 6 shows, the (absolute) interaction energy is highest for the [Keggin][Emim] system, which is in agreement with the diffusion and sorption results.

4. CONCLUDING REMARKS

In this study, we have performed MD simulations to investigate the separation of the CO_2/CH_4 gas mixture using $[Emim]_3$[Keggin] IL confined into (26,0) CNT at several external pressures at 300 K. We have also compared the results with the different ILs, including [Emim][BF_4], [Emim][PF_6], and [Emim][$AlCl_4$]. In another model, the membrane was constructed by four similar CNTs (with the confined ILs), which are almost spaced equally from each other. The most important results are as follows:

(1) In general, the sorption of CO_2 in the IL−CNT systems is higher compared to CH_4. This is due to the stronger interaction of CO_2 molecules with the CNT and IL.

(2) Carbon nanotubes with $[Emim]_3[PW_{12}O_{40}]$ IL exhibit the highest sorption of CO_2 (with the exception of highest pressure) owing to the high electric charge of this particular IL.

(3) Totally, the CNTs without IL exhibit higher sorption of CH_4 (especially in 4CNTs) since CO_2 with its smaller size has a higher passage rate compared to CH_4. For the same reason, the CNTs without IL exhibit smallest sorption of CO_2.

(4) It is also totally observed that $AlCl_4^-$ exhibits smaller sorption than the other ILs, especially for the 4ILs−4CNTs system for CO_2 sorption.

(5) Totally, the presence of the ILs in the 1CNT and 4CNTs configurations results in a greater CO_2 rejection rate, which is due to the IL−CO_2 interactions. With the exception of very low pressure (P = 5 bar), the rejection rate of $[Emim]_3[PW_{12}O_{40}]$ IL is higher than the other ILs.

(6) The nanocomposites show a higher rejection rate for CO_2 over CH_4 at low pressure ranges. The $[Emim]_3[PW_{12}O_{40}]$ IL in the 4ILs−4CNTs configuration exhibits more than 2 times rejection rate than the 4CNTs without IL, which approves the importance of polyoxometalate IL−CNT nanocomposites in the gas separation process.

(7) The diffusion coefficients of confined gas molecules in the CNT are smallest for the [Keggin][Emim] IL, which is in agreement with the sorption and rejection results and is due to the greater gas−[Keggin][Emim] IL interactions. Also, the diffusion coefficients of confined CO_2 molecules are smaller than those of the CH_4 molecules, which is due to the stronger CO_2−IL interactions than CH_4−IL interactions.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.iecr.3c01924.

Parameters of all interactions used in this work (Table S1) (PDF)

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Notes

The authors declare no competing financial interest.

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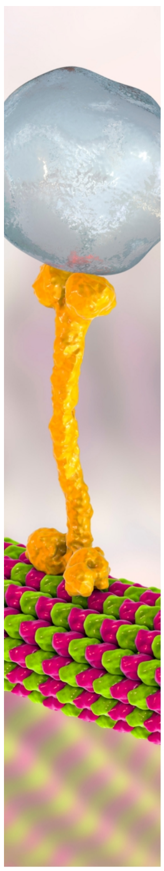

https://doi.org/10.1021/acs.iecr.3c01924
Ind. Eng. Chem. Res. 2023, 62, 14548−14556